M. JIJNASH KUMAR

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· LinkedIn: https://www.linkedin.com/in/jijnash-kumar-mukka/

· GitHub: https://github.com/Jijnash2636

Undergraduate researcher and third-year B.Tech student in Artificial Intelligence and Machine Learning (CGPA: 9.48/10), with a strong foundation in biomedical and neurological data analysis. Experienced in predictive modeling, simulation-based experimentation, and explainable AI for real-world applications. Currently expanding research into AI-driven smart systems, including carbon emission control and traffic optimization, while continuing work on multimodal healthcare diagnostics and Lab-on-Chip-based biomarker analysis. Seeking research opportunities in interdisciplinary AI applications across healthcare, sustainability, and intelligent systems.

RESEARCH INTERESTS

- Biomedical AI
- Multimodal Diagnostics
- Explainable AI (XAI)
- Clinical Decision Systems
- Predictive Modelling

EDUCATION

B.Tech, Artificial intelligence & machine learning (2023 – Present)
SRM Institute of Science and Technology, Tiruchirappalli, CGPA – 9.48/10
Graduation year (2027)

12th (Senior Secondary Examination) 2022
STATE Board, 76.8%

10th (Secondary Examination) 2020
STATE Board, 99%

Research Experience

IIIT Nagpur – Research Intern (CSE)
Dec 2025 – Feb 2026
• Developed multimodal ML models for Parkinson's disease detection using voice, facial, and hand biomarkers
• Designed pipelines for non-imaging biomedical data
• Explored feature fusion techniques for improved prediction
• Evaluated models using standard classification metrics

Research & Projects

EcoSignal-CP: Predictive Carbon Emission Control
- Built CO_2 prediction model using SUMO; reduced peak emissions by ~11.8%
- Applied simulation-based optimization

AI-Based Clinical Blood Analysis (CBC)
- Developed ML model for disease prediction using hematological data
- Performed feature engineering and evaluation (ROC-AUC)

Multimodal Parkinson's Detection System
- Developed ML model using voice, facial, and motor biomarkers
- Focused on early, non-invasive detection

Emergency Brain 360 – AI Clinical Triage (Top 6 National)

- Built AI-assisted triage system with real-time dashboard

Hospital Analytics Platform (MedAIthon – 5th Place)

- Designed system for patient flow optimization

Technical Skills

Python, C++

Machine Learning, Feature Engineering, Explainable AI

Scikit-learn, Pandas, NumPy

SUMO, FastAPI, Google Cloud (Vertex AI)

Research & Publications

Research Paper (In Progress): *EcoSignal-CP: Carbon Pressure Control for Urban CO_2 Emission Stabilization*

Book Chapter Publication – The Dawn Publishers (2024)

Internships & Training

Generative AI Internship - Google Cloud (AICTE & EduSkills)
AI & Machine Learning Internship - AWS Academy
ServiceNow AI Automation Internship - AICTE & SmartBridge
AI Agent for Healthcare Diagnostics - Infosys Springboard

Achievements

- Top 6 National Rank - Smartathon 2025
- 5th Place & Special Prize – MedAIthon 2025
- CGPA 9.48/10 – Consistent Academic Excellence

CERTIFICATIONS

- **RHCSA** (Red Hat Certified System Administrator)
- Google Cloud GenAI | AWS ML | ServiceNow AI | Infosys AI

Professional Objective

To pursue advanced research in AI-driven interdisciplinary systems, contributing to scalable solutions in healthcare diagnostics, smart urban infrastructure, and carbon emission control, while preparing for graduate-level research and innovation.

REFERENCES

Dr. Jitendra V. Tembhurne
Head of the Department & Assistant Professor
Department of Computer Science & Engineering
Indian Institute of Information Technology, Nagpur
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Dr. Balaji Ganesh Rajagopal
Associate Professor & Head of the Department (AIML)
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